

Total No. of Questions : 6]

SEAT No. :

P175

[Total No. of Pages : 2

OCT/BE/Insem.-104
B.E.(IT)
MACHINE LEARNING
(2012 Pattern) (Semester - I)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates :

- 1) *Draw neat diagrams wherever necessary.*
- 2) *Assume suitable data, if necessary.*
- 3) *Figures to the right indicate full marks.*

Q1) a) Explain supervised learning, unsupervised learning and semi-supervised learning. **[6]**

b) Write a note on features. **[4]**

OR

Q2) a) Explain, in brief, geometric models. **[6]**

b) What is Machine Learning? Give an example. **[4]**

Q3) a) Calculate TP, TN, true negative rate (tnr), accuracy and pos for the following : **[5]**

	Predicted +	Predicted –
Actual +	50	25
Actual –	5	20

b) Compare 1-vs-1 and 1-vs-rest. **[5]**

OR

Q4) a) Prove: error rate = 1-accuracy. **[5]**

b) Explain bias-variance dilemma. **[5]**

P.T.O.

- Q5) a)** What is lasso and ridge regression? [5]
b) Explain perceptron. [5]

OR

- Q6) a)** Explain slack variables, hard margin and soft margin. [5]
b) Explain sigmoid function. [5]

